

7 (a) (i) Number of protons = 57, number of neutrons = $(141 - 57) = 84$ [B1]

(ii) Mass defect, $\Delta m = [(141 - 57)(1.009) + 57(1.007) - 140.911]$ [M1]

$$= 1.244(1.66 \times 10^{-27})$$

$$= 2.06504 \times 10^{-27} \text{ kg}$$

$$\text{Binding energy} = \Delta mc^2$$

$$= (2.06504 \times 10^{-27})(3.00 \times 10^8)^2 \quad [\text{M1}]$$

$$= 1.858536 \times 10^{-10} \text{ J} \quad [\text{M1}]$$

$$= 1161.585 \text{ MeV}$$

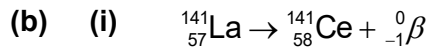
$$= 1162 \text{ MeV} \quad [\text{A0}]$$

(iii) B.E. per nucleon of Pu-239 = $\frac{1807}{239}$
= 7.56 MeV

B.E. per nucleon of La-141 = $\frac{1162}{141}$
= 8.24 MeV [M1: for both energies]

Since plutonium-239 nucleus has a lower binding energy per nucleon, it is less stable against fission [B1]

and is more likely to undergo nuclear fission. [A1]



Electron or beta particle [B1]

(ii) There is emission of a neutrino (or anti-neutrino, or third particle) along with an electron. [M1]

The energy/momentum is shared between the electron and neutrino (anti-neutrino) in various ways (or such that the neutrino also has a range of



energy/momentum), while maintaining conservation of energy and momentum. [A1]

Hence the electrons and neutrinos are emitted with a range of energies.

